$|g_{a\gamma}|$ [GeV^{-1}]

m_a [eV]

CAST

Solar ν

Globular clusters

White dwarfs

SN1987A (ν)

Diffuse-γ

PVO

GW170817

GW170817 (γ)

SN1987A (γ)

BBN

StAB

Solar basin

M82

Freeze-in

$\tau_{a\to\gamma\gamma} < t_U$

HST

WINE RED

JWST

MUSE

DESI

HST

Ionisation fraction

CMB

Leo T

Cosmic background

X-rays